Case Study: Ahmedabad Heat Action Plan (HAP)

Pioneer Heat Resilience Framework and Urban Adaptation Blueprint in the Global South

Policy Analysis Report

July 30, 2026

Abstract

This case study evaluates the development, operational mechanics, quantitative health outcomes, and scalability of the Ahmedabad Heat Action Plan (HAP). Formulated in 2013 following a deadly 2010 heatwave, the plan was South Asia's first comprehensive municipal heat early warning and disaster response system. By blending real-time meteorological alerts, public education, healthcare system capacity-building, and high-albedo structural interventions (*Cool Roofs Program*), the Ahmedabad HAP has prevented an estimated 1,100 to 1,200 deaths annually and now serves as the model framework across India and the Global South.

1 Background & Catalyst Event

In May 2010, the city of Ahmedabad in Gujarat, India, experienced a historic heatwave event, with ambient temperatures reaching an extreme peak of 46.8°C (116.2°F).

1.1 Public Health Impact

- **Excess Mortality:** Over $1,344$ excess deaths were registered in May 2010 alone, representing a 43% spike in all-cause mortality relative to baseline years.
- **Healthcare Overload:** Emergency rooms and intensive care units were overwhelmed with acute heat-related illnesses (HRI), while public awareness regarding heat stress risks remained critically low.

In response to this crisis, the **Ahmedabad Municipal Corporation (AMC)** established a public health partnership with the **Indian Institute of Public Health, Gandhinagar (IIPHG)**, the **Natural Resources Defense Council (NRDC)**, and international technical partners to engineer South Asia's first dedicated Heat Action Plan, launched in **2013**.

2 The Four Pillars of the Ahmedabad HAP

The plan coordinates emergency response actions across multiple city departments via four fundamental pillars:

2.1 Pillar 1: Early Warning System (EWS) & Inter-Agency Cascades

The India Meteorological Department (IMD) transmits 7-day probabilistic temperature forecasts to the AMC Nodal Officer. Forecasts are categorized into actionable risk thresholds:

- **Yellow Alert (**41.0°**C –** 43.0°**C):** Hot Day Advisory.
- **Orange Alert (**43.1°**C –** 44.9°**C):** Severe Heat Alert.

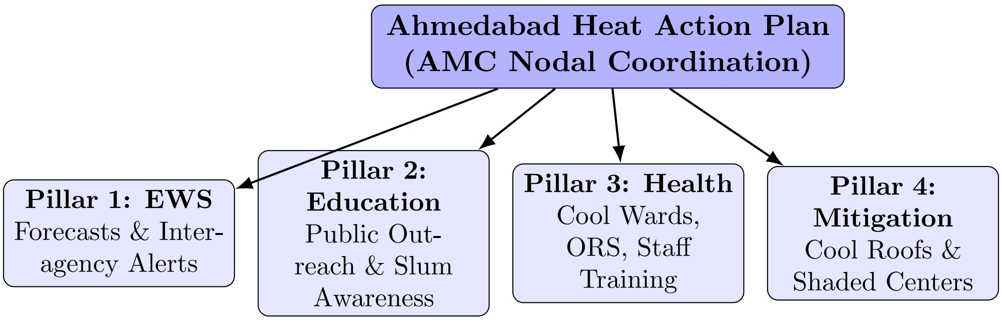

Figure 1: Systemic Architecture of the Ahmedabad Heat Action Plan.

- **Red Alert ($\geq 45.0°$C):** Extreme Heat Alert (Emergency Response).

Upon alert generation, emergency operational procedures trigger across traffic police, public transport (AMTS/BRTS), water supply divisions, and healthcare centers.

2.2 Pillar 2: Public Education and Vulnerable Community Outreach

- **Mass Communication:** Alerts display across electronic LED panels at major traffic intersections, sent via automated SMS/WhatsApp messages, and broadcast via local radio and print media.
- **Field-Level Outreach:** Accredited Social Health Activists (ASHA) and Urban Health Center link workers distribute pamphlets and conduct door-to-door educational campaigns in high-risk informal settlements (*slums*).

2.3 Pillar 3: Healthcare System Capacity Building

- **Cool Wards:** Dedicated climate-controlled wards are reserved in major municipal hospitals (e.g., VS Hospital, LG Hospital, Shardaben Hospital).
- **Medical Logistics:** Strategic stockpiling of Oral Rehydration Salts (ORS), intravenous (IV) fluids, and ice packs across Urban Health Centers (UHCs).
- **Emergency Training:** Paramedics and 108 ambulance personnel receive specialized training in early recognition and rapid immersion/cooling treatment for exertional heatstroke.

2.4 Pillar 4: Adaptive Structural Mitigation & Passive Cooling

- **Cooling Havens:** Public spaces, temples, libraries, and night shelters remain open during peak sunlight hours (12:00 PM–4:00 PM) to offer public heat respite.
- **Traffic Intersection Management:** Traffic lights at major intersections are switched to flashing yellow during extreme peak afternoon hours to prevent commuters from idling in direct solar heat.
- **Ahmedabad Cool Roofs Program:** Deployment of high-albedo white paint, cool tiles, or lime washes on rooftops. The passive energy balance formula highlights the thermal reduction:

$$Q_{\text{absorbed}} = (1 - \alpha) \cdot I_{\text{solar}} \tag{1}$$

Where increasing the roof surface albedo α lowers solar energy absorption, reducing indoor residential temperatures by 2°C to 5°C without active electrical power consumption.

3 Seasonal Operational Workflow

The operational architecture runs along a three-phase calendar cycle:

Table 1: Seasonal Phased Implementation of the Ahmedabad HAP

Phase	Timeline	Primary Action Items
Phase I: Pre-Heat	Jan – Mar	Inter-agency alignment meetings; updating informal settlement vulnerability maps; healthcare inventory check; launching public awareness drives.
Phase II: Active Heat	Apr – Jun	Monitoring daily IMD temperature forecasts; issuing color-coded alerts; opening public cooling shelters; track hospital HRI admissions; enforcing water distribution schedules.
Phase III: Post-Heat	Jul – Sep	Conducting mortality and morbidity evaluations; auditing departmental responses; updating the plan for the next seasonal cycle.

4 Quantifiable Public Health Outcomes

Rigorous epidemiological evaluations published in leading public health journals demonstrate the significant life-saving impact of the HAP:

Table 2: Comparative Impact Assessment Before and After HAP Implementation

Indicator	Pre-HAP Baseline (2010)	Post-HAP Evaluation
Annual Mortality	+43% excess mortality in May 2010 ($> 1,300$ deaths)	$1,100 - 1,190+$ **deaths avoided** annually across the municipality
Risk at $47°$**C**	Mortality relative risk increased $> 100\%$	Relative risk increase capped at $\approx 25\%$
Heatstroke Fatalities	65 registered direct heatstroke deaths (2010)	Reduced to < 15 **cases annually** on average

5 National Legacy and Policy Challenges

5.1 Scaling and Replicability

The success of the Ahmedabad plan catalyzed a national paradigm shift in heat disaster management:

1. **National Blueprint:** India's National Disaster Management Authority (NDMA) adopted the Ahmedabad model as the national policy template.
2. **Widespread Adoption:** Over 23 heat-prone Indian states and more than 100 urban centers have implemented localized Heat Action Plans adapted from this initial framework.

5.2 Remaining Operational Challenges

- **Informal Labor Protections:** Enforcing rest intervals and shift adjustments for informal daily wage laborers, street vendors, and construction workers remains difficult due to economic pressure.
- **Urban Heat Island (UHI) Escalation:** Rapid concrete construction and loss of green canopy continue to elevate nocturnal temperatures, requiring expanded urban forestry programs alongside cool roof mandates.

6 Conclusion

The Ahmedabad Heat Action Plan serves as a benchmark for municipal climate adaptation in low- and middle-income countries. By establishing early warnings, cross-departmental coordination, medical readiness, and passive cooling strategies, the initiative proves that targeted, low-cost public health interventions can deliver immediate, measurable resilience against rising extreme heat events.